CVR COLLEGE OF ENGINEERING

Department of CSE

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NOTO

A Web-Based Thinking,Note making and Collaboration Ecosystem

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INTRODUCTION

Problem Domain

- Learning and collaboration today happen across fragmented tools — whiteboards, note-taking apps, document viewers, and messaging platforms.
- This fragmentation disrupts thinking flow, reduces clarity, and increases cognitive load for users.

Project Objective

- Noto provides a single unified digital space where users can think, learn, collaborate, and retain knowledge effectively.
- Built around a shared infinite canvas with an embedded AI assistance layer for seamless workflow integration.

PROBLEM DEFINITION

Core Issue: Existing digital tools focus on either collaboration or note-taking, but rarely both in an integrated and context-aware manner.

Key Problems Identified

01
No unified platform for visual thinking combined with structured notes

02
Poor integration between documents and freeform thinking spaces

03
Limited real-time collaboration in learning-focused tools

04
AI tools exist separately — not embedded into the thinking workflow

EXISTING WORK AND METHODS

1. Online Whiteboards	Enable real-time collaborative drawing and brainstorming but lack structured note-taking or document integration.
2. Note-Taking Applications	Support personal study and organization but are primarily single-user with limited freeform or visual capabilities.
3. Document Readers	Provide basic annotation features on PDFs but do not support dynamic collaboration or AI-driven assistance.
4. Standalone AI Tools	Offer summarization and explanation features but operate outside the user's core workspace, breaking context.

Critical Gap: These tools work independently, forcing users to switch contexts frequently and disrupting the natural flow of thinking.

CHALLENGES IN EXISTING SYSTEMS

- Fragmented user experience across multiple tools with no unified interface
- Lack of deep integration between document viewers and personal note-taking
- Collaboration tools are not optimized for structured learning environments
- AI assistance is not context-aware or embedded within the thinking workspace
- High cognitive overload from managing and switching between multiple platforms

LITERATURE REVIEW

Ref [1]	Online collaborative tools improve group learning outcomes but lack personalization and individual context retention.
Ref [2]	AI-based summarization techniques significantly improve learning retention and knowledge recall.
Ref [3]	Visual note-taking methods enhance conceptual understanding and long-term memory.
Ref [4]	Real-time collaboration systems measurably improve student engagement and participation.
Ref [5]	Integrated learning environments reduce cognitive load and improve task completion rates.

Conclusion: These studies collectively motivate the need for a unified, AI-assisted thinking platform that combines benefits of all reviewed approaches.

PROPOSED SOLUTION OVERVIEW

Noto: Core Architecture

A web-based thinking and collaboration ecosystem built around three foundational components:

1 Shared Infinite Canvas Engine

2 Common AI Intelligence Layer

3 Multiple Context-Aware Usage Modes

System Adaptability

The platform adapts to three primary user contexts, enabling a consistent experience across varied workflows:

Collaborative Learning

Multi-user shared canvas with live synchronization

Individual Study

Document annotation with AI-based explanations

Personal Thinking

Freeform planning, mind mapping, and note retention

PROPOSED METHOD

Technical Approach

- A single canvas engine reused and shared across all application modes
- A mode-based architecture that enables or disables features based on context
- A backend-mediated AI layer for consistent, centralized intelligence
- A client–server architecture with real-time synchronization support

Key Benefits

Scalability

Canvas engine scales to handle many concurrent users

Modularity

Each component is independently deployable and testable

Clarity

Separation of concerns keeps system design maintainable

SYSTEM MODULES

01
User Authentication

Manages user registration, secure login, and session handling across all modes.

02
Canvas Management

Handles infinite canvas creation, persistent storage, and retrieval per user.

03
Collaboration & Real-Time

Enables multi-user synchronization, live cursors, and instant updates via WebSockets.

04
Document Annotation

Supports PDF and document upload with inline annotation and markup tools.

05
AI Assistance

Provides on-demand summarization, concept explanation, and mind-map generation.

06
Data Persistence

Manages all database operations, ensuring data integrity and reliable recovery.

Design Principle: Each module is loosely coupled and independently extendable.

FUNCTIONAL REQUIREMENTS

01 **User Registration & Login**

Secure authentication with session management and role-based access control.

02 **Board Creation & Management**

Create, edit, delete, and organize multiple boards within a personal workspace.

03 **Infinite Canvas & Drawing**

Freeform sketching, annotation, and sticky-note tools on an unbounded canvas.

04 **Real-Time Collaboration**

Simultaneous multi-user editing and viewing with live cursor presence.

05 **Document Upload & Annotation**

Upload PDF and document formats; annotate and link notes directly to content.

06 **AI Summarization & Explanation**

On-demand AI processing to summarize, explain, and generate mind maps.

07 **Auto-Save & Data Retrieval**

Automatic persistence of all canvas and note states with reliable recovery.

NON-FUNCTIONAL REQUIREMENTS

Scalability

Architecture supports multiple concurrent users without performance degradation.

Low Latency

Real-time collaboration features maintain minimal delay during active editing sessions.

Data Security

Robust access control, input validation, and encrypted data storage and transit.

High Availability

System designed for reliable uptime with fault tolerance and graceful error recovery.

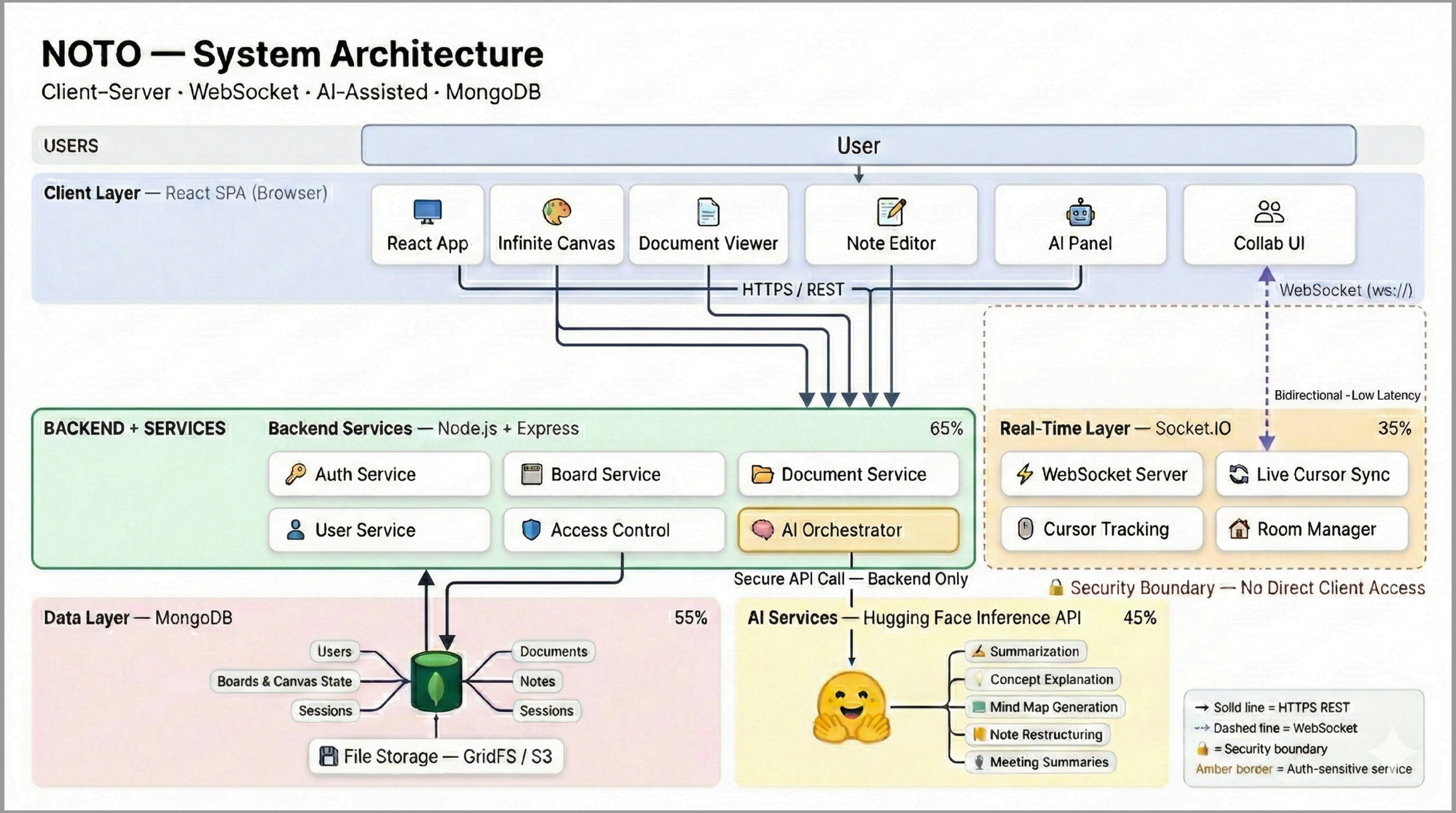
Intuitive UI

Minimal learning curve through clean interface design and consistent interaction patterns.

Cross-Platform

Fully functional across all modern web browsers with no platform-specific dependencies.

ARCHITECTURE DIAGRAM



HARDWARE REQUIREMENTS

Minimum 4 GB RAM

Ensures smooth client-side rendering and canvas operation without performance issues.

Standard Desktop or Laptop

No specialized or high-performance hardware is required to run the application.

Internet Connectivity

Required for real-time collaboration features, AI assistance, and cloud storage sync.

Note: The system is designed to be lightweight and accessible, requiring only standard computing equipment available to most users — no specialized hardware is needed.

SOFTWARE REQUIREMENTS

Frontend

React

HTML5 Canvas API

Backend

Node.js

Express.js

Database

MongoDB

Real-Time

Socket.IO

AI Layer

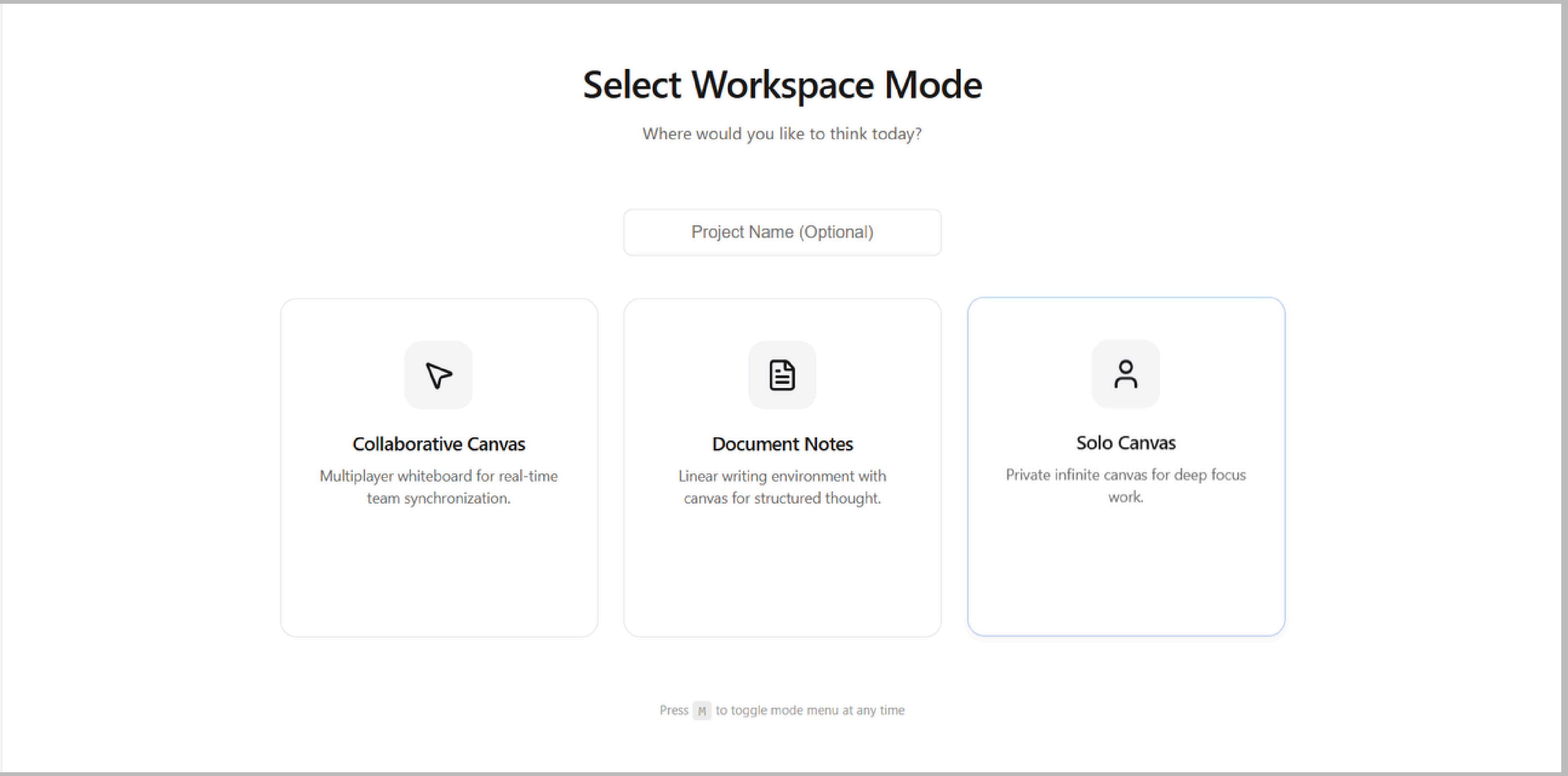
Hugging Face Inference API

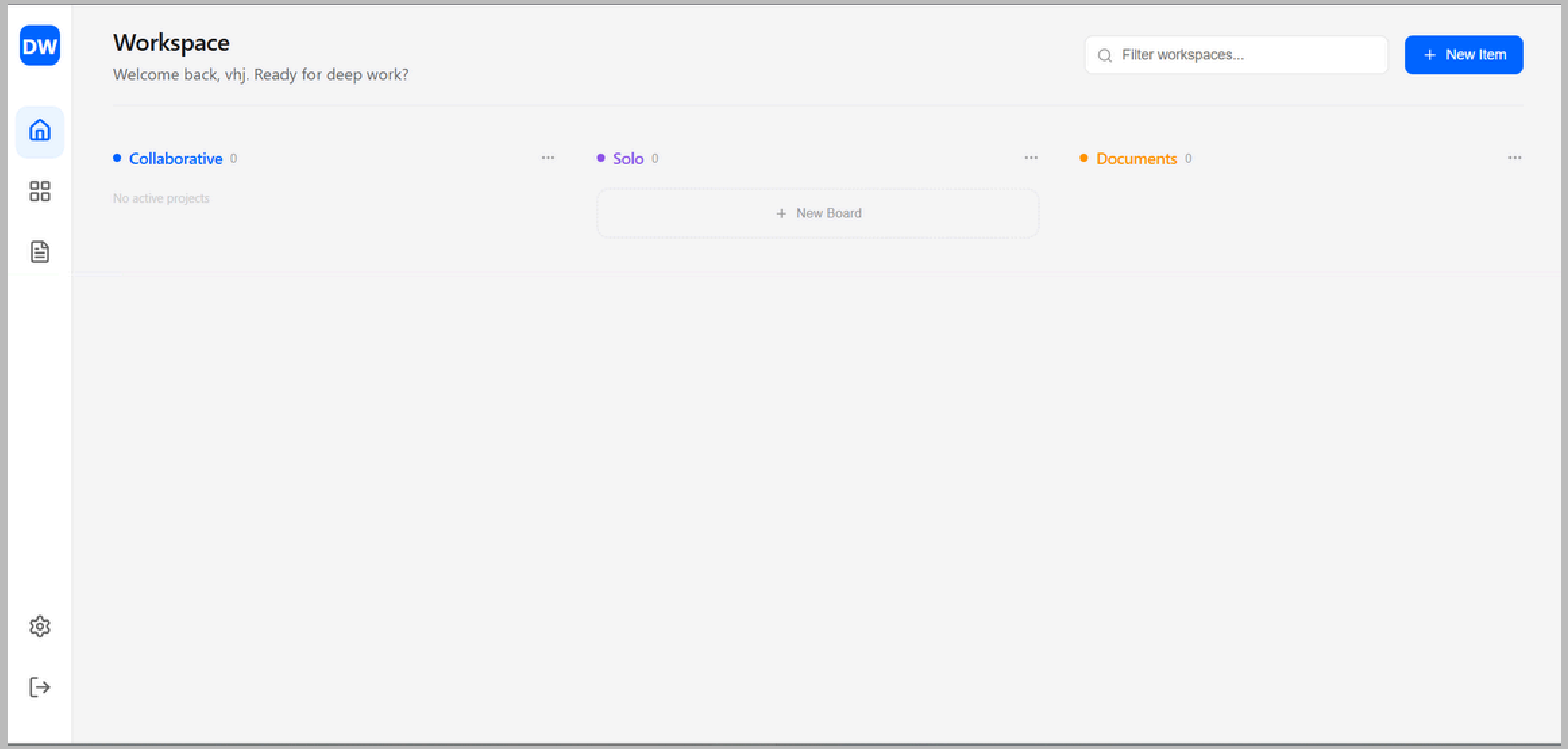
Dev Tools

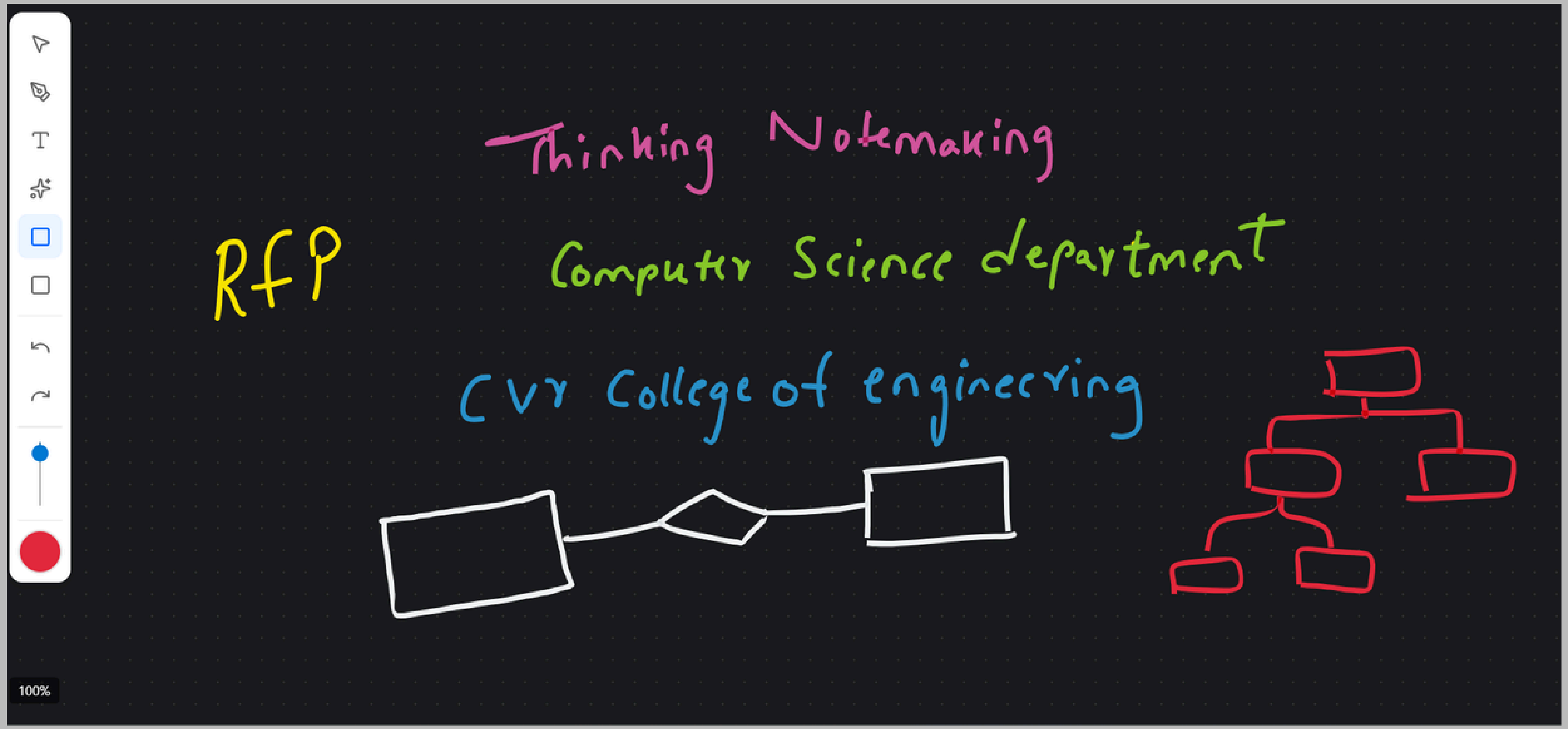
Git

Anti Gravity

IMPLEMENTATION



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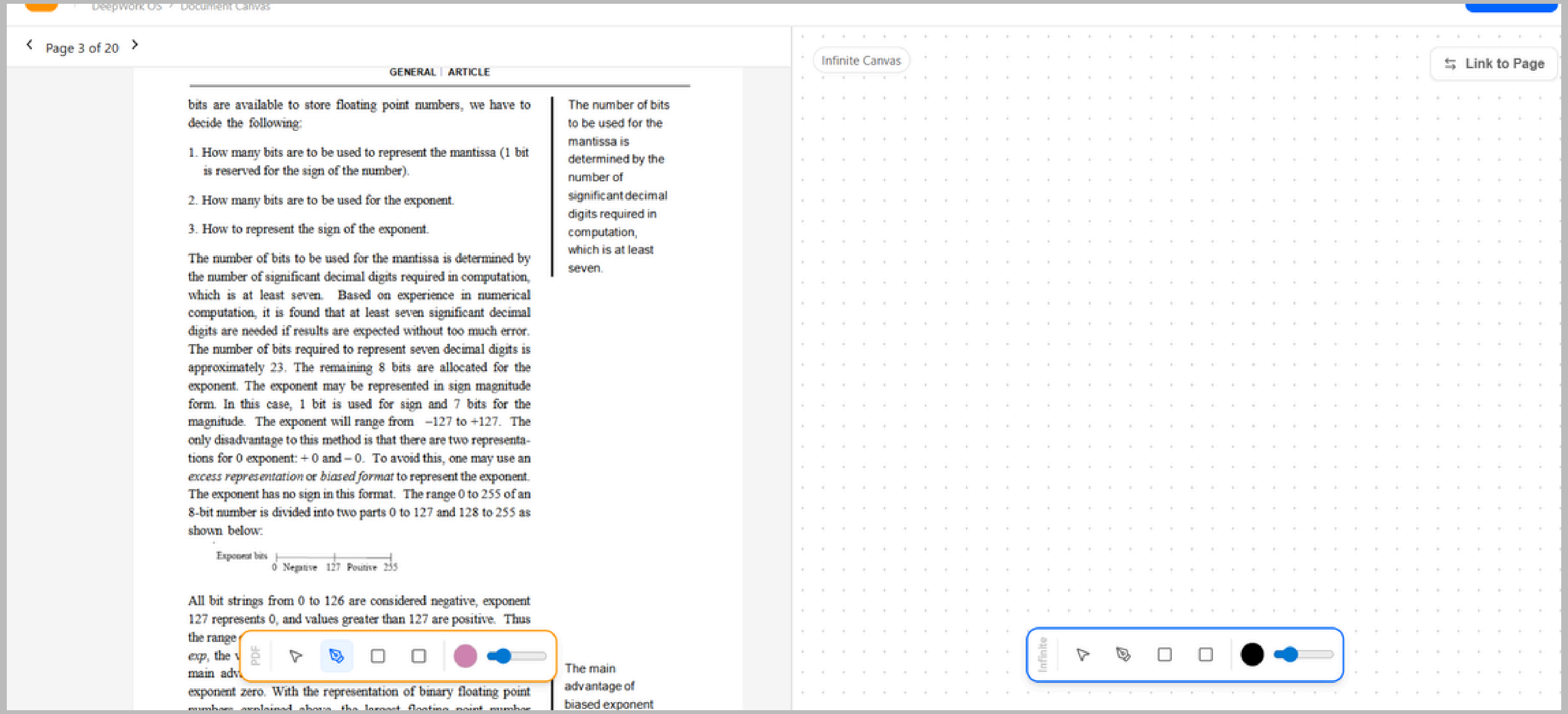
DeepWork OS / Document Canvas
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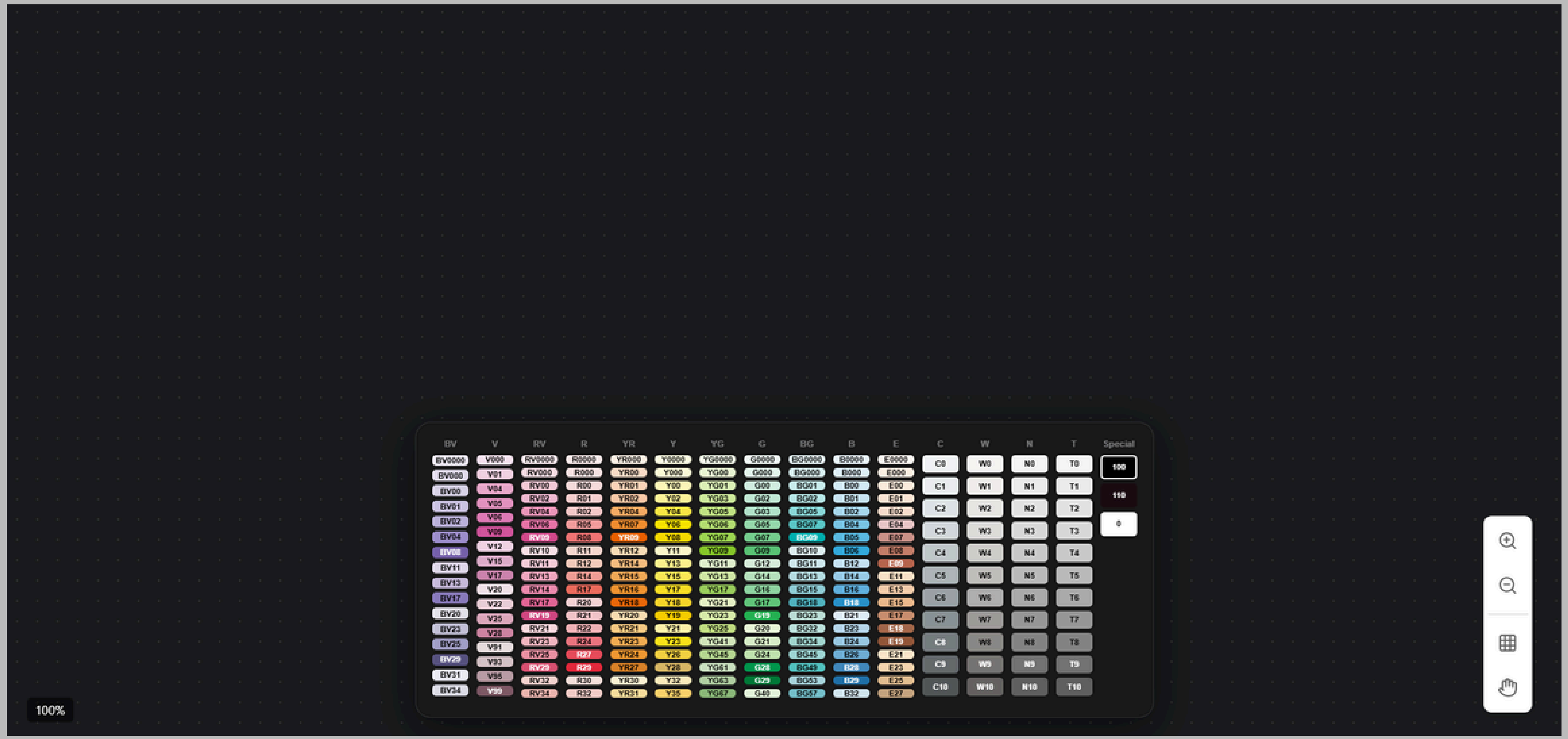
GENERAL | ARTICLE
bits are available to store floating point numbers, we have to decide the following:
1. How many bits are to be used to represent the mantissa (1 bit is reserved for the sign of the number).
2. How many bits are to be used for the exponent.
3. How to represent the sign of the exponent.
The number of bits to be used for the mantissa is determined by the number of significant decimal digits required in computation, which is at least seven. Based on experience in numerical computation, it is found that at least seven significant decimal digits are needed if results are expected without too much error. The number of bits required to represent seven decimal digits is approximately 23. The remaining 8 bits are allocated for the exponent. The exponent may be represented in sign magnitude form. In this case, 1 bit is used for sign and 7 bits for the magnitude. The exponent will range from −127 to +127. The only disadvantage to this method is that there are two representations for 0 exponent: + 0 and – 0. To avoid this, one may use an excess representation or biased format to represent the exponent. The exponent has no sign in this format. The range 0 to 255 of an 8-bit number is divided into two parts 0 to 127 and 128 to 255 as shown below:
Exponent bits
0 Negative 127 Positive 255
All bit strings from 0 to 126 are considered negative, exponent 127 represents 0, and values greater than 127 are positive. Thus
exponent zero. With the representation of binary floating point
The number of bits to be used for the mantissa is determined by the number of significant decimal digits required in computation, which is at least seven.
The main advantage of biased exponent
PDF
Infinite Canvas
Link to Page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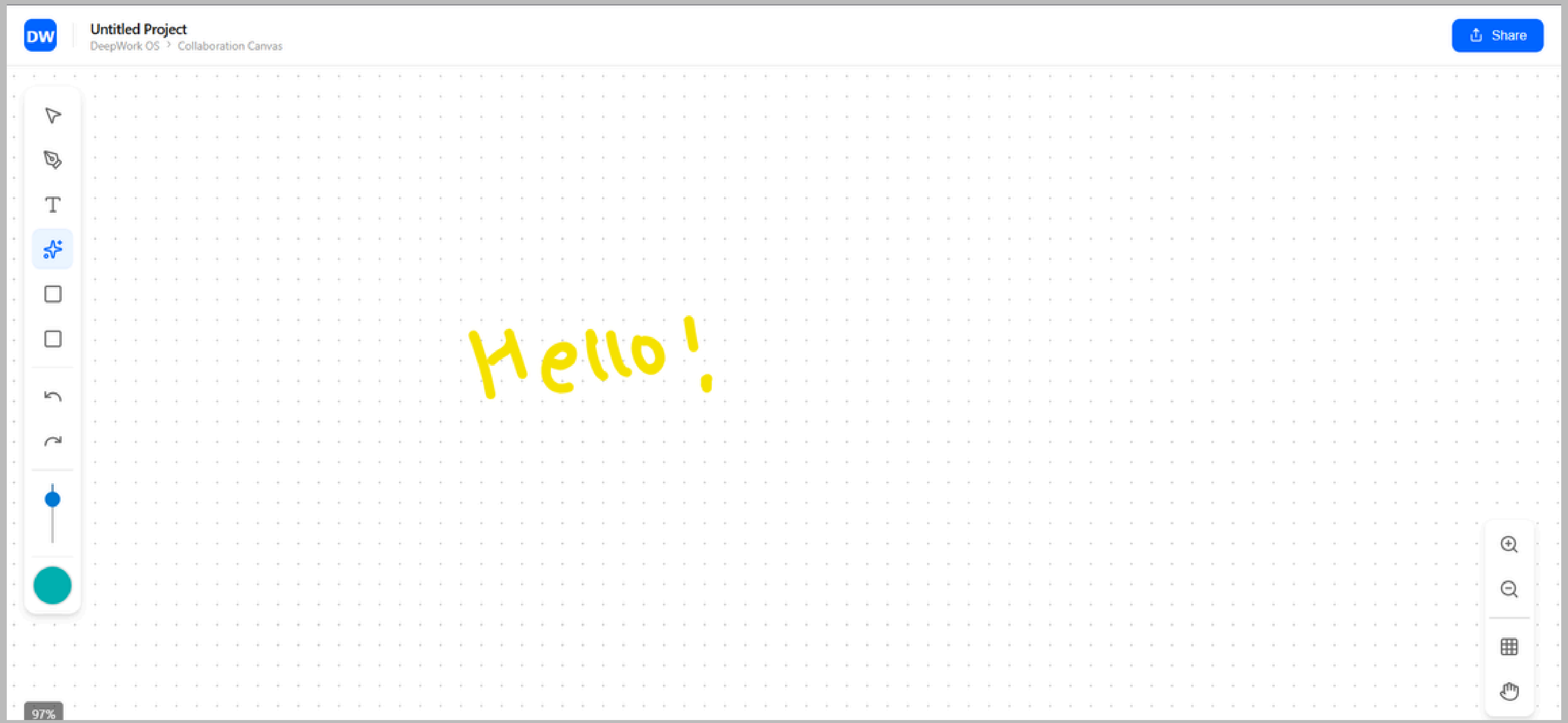
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DeepWork OS > Collaboration Canvas
Share
Hello!
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CONCLUSION

Problem Addressed

Noto addresses the fragmentation problem by providing a unified platform that combines an infinite canvas, document-based notes, and AI assistance in one seamless workspace.

Technical Approach

The modular architecture ensures scalability and maintainability, with loosely coupled components designed to be independently extended or replaced.

System Capabilities

The platform supports multiple usage modes — collaborative learning, individual study, and personal thinking — adapting to the diverse needs of its users.